

3.5.2 Selection Statements (if, if-else, if-elif-else, nested)

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 10 of 13

★ Golden

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LEARN IT

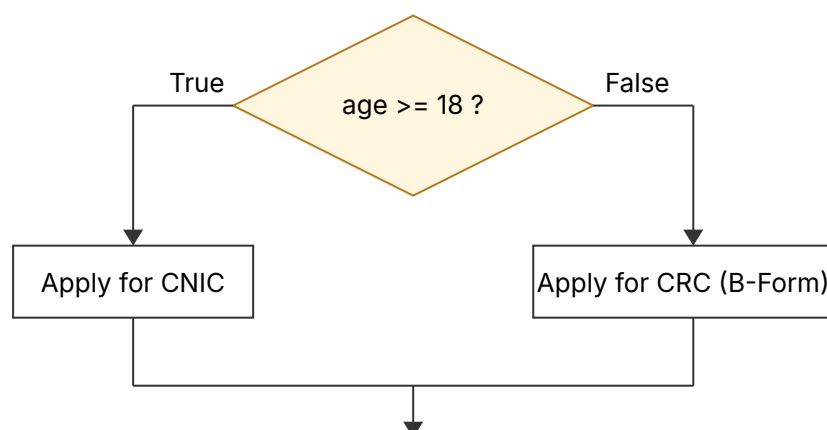
Selection changes the flow of a program based on a **condition** — an expression that is either True or False. It helps a program respond intelligently to different situations.

1. The if statement executes a block of code **only if** the condition is True. If the condition is False, the block is completely skipped.

```
if condition:
    statement(s)    # runs only when condition is True
```

2. The if-else statement provides **two paths**: if the condition is true, do this; **else** (otherwise), do that. Exactly one of the two blocks always runs.

```
if condition:
    block A        # condition True
else:
    block B        # condition False
```



An if-else statement: one diamond (condition), two paths, then the paths join again.

WORKED EXAMPLE

The `if` statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("You are an Adult.")    # runs only if age ≥ 18
print("Program continues...")    # runs every time
```

Run 1 (user types 20):

```
OUTPUT
Enter your age: 20
You are an Adult.
Program continues...
```

Run 2 (user types 15):

```
OUTPUT
Enter your age: 15
Program continues...
```

The last line is not indented, so it is outside the `if` block and runs every time.

The `if-else` statement.

```
age = int(input("Enter your age: "))
if age ≥ 18:
    print("Apply for CNIC.")
else:
    print("Apply for CRC (B-Form).")
```

Sample run (user types 16):

```
OUTPUT
Enter your age: 16
Apply for CRC (B-Form).
```

Even or odd — a classic `if-else` program:

```
n = 7
if n % 2 == 0:
    print(n, "is even")
else:
    print(n, "is odd")
```

```
OUTPUT
7 is odd
```

$7 \% 2 = 1$, and $1 == 0$ is `False`, so the `else` block runs.

REMEMBER IT

Formatting rule: put a **colon (:)** after `if` and `else`, and **indent** the code inside the blocks. `else` never has a condition of its own.

COMMON MISTAKE

Common mistakes: forgetting the colon (`SyntaxError: expected ':'`), writing a condition after `else` (`else age < 18:` is wrong), and not indenting `else` at the same level as its `if`.

اردو میں سمجھیں

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔ `if` کا بلاک صرف تب چلتا ہے جب شرط درست (True) ہو۔ `if-else` میں دو راستے ہوتے ہیں: شرط درست ہو تو پہلا بلاک، ورنہ `else` والا بلاک چلتا ہے۔ `if` اور `else` کے بعد کولن (:) اور اندر کے کوڈ کی انڈینٹیشن ضروری ہے۔

EXAM TIP

Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`, the condition, both blocks, and a sample run.

CHECK YOURSELF

1. Predict the output when `marks = 32`:

```
marks = 32
if marks ≥ 33:
    print("Pass")
else:
    print("Fail")
print("Done")
```

2. Find the two errors:

```
n = 5
if n > 0
    print("Positive")
else n ≤ 0:
    print("Not positive")
```

Answers: 1. Fail then Done. 2. Missing colon after `if n > 0`; `else` must not have a condition — write `else:`.

LEARN IT

3. The `if-elif-else` statement handles **many conditions**. (`elif` is short for "else if".) Python checks the conditions **from top to bottom**. As soon as it finds a True condition, it runs that block and **ignores all the rest**. If none is True, the `else` block runs.

4. Nested `if` is an `if` inside another `if`. It is used when a decision needs to pass **several levels of checks**: the inner check happens only if the outer check passed.

WORKED EXAMPLE

Grading with if-elif-else.

```
marks = int(input("Enter marks: "))
if marks ≥ 80:
    print("A-1 Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 60:
    print("B Grade")
else:
    print("Fail or below B")
```

Sample run (user types 75):

OUTPUT

```
Enter marks: 75
A Grade
```

Trace for different marks:

marks	≥ 80?	≥ 70?	≥ 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Loan approval with nested if.

```
age = 20
salary = 35000
if age ≥ 18:
    # passed first check
    if salary ≥ 30000:
        # passed second check
        print("Loan APPROVED!")
    else:
        print("Salary too low for loan.")
else:
    print("Too young for loan.")
```

OUTPUT

```
Loan APPROVED!
```

age	salary	Outer check (age ≥ 18)	Inner check (salary ≥ 30000)	Output
20	35000	True	True	Loan APPROVED!
20	25000	True	False	Salary too low for loan.
16	35000	False	(never checked)	Too young for loan.

When you only need a yes/no answer, two levels can often be joined with `and` — but then you lose the separate messages:

```
age = 20
salary = 25000
if age ≥ 18 and salary ≥ 30000:
    print("Loan APPROVED!")
else:
    print("Loan not approved.")
```

OUTPUT

Loan not approved.

REMEMBER IT

In an `if-elif-else` ladder, put the **strictest condition first** (highest marks first). Think of a ladder: you step off at the first rung that fits.

COMMON MISTAKE

Wrong order gives wrong answers. Here 85 marks gets "B Grade", because `85 ≥ 60` is True first and the other conditions are ignored:

```
marks = 85
if marks ≥ 60:
    print("B Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 80:
    print("A-1 Grade")
```

OUTPUT

B Grade

This is a **logical error**: no crash, just a wrong result. Fix it by checking `≥ 80` first.

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔ نیسٹڈ ایف (Nested if) میں ایک if دوسرے if کے اندر ہوتا ہے، اور اندر والی شرط تب ہی دیکھی جاتی ہے جب باہر والی شرط درست ہو۔

EXAM TIP

Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks carefully (8 spaces for the inner block), and give a sample run. Tracing tables like the ones above earn full marks in "predict the output" questions.

CHECK YOURSELF

1. Predict the output:

```
temp = 30
if temp > 35:
    print("Hot")
elif temp > 25:
    print("Warm")
elif temp > 15:
    print("Pleasant")
else:
    print("Cold")
```

2. Predict the output:

```
x = 15
if x > 10:
    if x > 20:
        print("A")
    else:
        print("B")
else:
    print("C")
```

Answers: 1. Warm (30 > 35 is False; 30 > 25 is True, so the rest are skipped even though 30 > 15 is also True). 2. B (outer check 15 > 10 passes; inner check 15 > 20 fails).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. if runs its block when the condition is:

- a. True
- b. False
- c. A string always
- d. A list

MCQ 2. else runs when:

- a. The if condition is False
- b. Always
- c. Never
- d. Only on Sundays

MCQ 3. elif is for:

- a. More conditions, first True wins
- b. Loops
- c. Files
- d. Imports

MCQ 4. Nested if means:

- a. An if inside another if
- b. Two prints
- c. A stack pop
- d. A pie

MCQ 5. marks $\geq$ 40 is a:

- a. Boolean condition
- b. Assignment
- c. Comment
- d. Gate delay

MCQ 6. if x=5: is:

- a. Syntax error (use ==)
- b. Correct assignment if
- c. A loop
- d. A class

Extra short questions

1. Write if-else to print Pass if marks $\geq$ 40.
2. When do you need elif vs nested if?
3. What is a dead elif?
4. Trace marks=75 through A $\geq$ 80, B $\geq$ 65, C $\geq$ 50, else.

Extra long question

1. Write a nested-if program: if present, then if marks $\geq$ 40 Pass else Fail; if absent print Absent. Show traces for three students.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. `if marks >= 40: print("Pass") else: print("Fail")`.
2. `elif`: mutually exclusive bands (A/B/C). Nested: a second question that only matters if the first is True.
3. A later condition that can never run because an earlier one already caught those values.
4. Not A, B True → B. Later `elif`s skipped.

Long

1. Outer: present yes/no. Inner: pass/fail. Traces: Absent; Present 30 Fail; Present 55 Pass. Indentation must be correct.

Next lecture: 3.5.3 Repetition Statements (for loop, while loop)